MeDeA: Decomposable Attention for Intrinsically Interpretable ECG Diagnosis

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ABSTRACT

Clinical adoption of automated ECG diagnosis is hindered by the opacity of deep learning models and the "winner-takes-all" feature collapse in standard attention mechanisms. To resolve these limitations, we propose **MeDeA (Multi-disease Decompositional Attention)**, a novel intrinsic interpretable architecture that reframes multi-label classification as a set of parallel, decomposed sub-problems. By assigning independent **Multi-Query Attention Heads (MQAH)** to distinct cardiac conditions, MeDeA effectively captures diverse, multi-focal morphological evidence, avoiding the explanation collapse of conventional methods. Experiments on the PTB-XL dataset demonstrate that MeDeA achieves a Macro-F1 of **0.775**, matching state-of-the-art black-box baselines while significantly improving attribution faithfulness. By aligning model reasoning with clinical features, MeDeA offers a transparent solution for trustworthy human-AI collaboration in cardiology.

1. Introduction

The electrocardiogram (ECG) is a cornerstone of cardiovascular diagnostics, providing a non-invasive and cost-effective means to monitor cardiac electrical activity. Each year, hundreds of millions of ECGs are recorded worldwide [1], underscoring its indispensable role in detecting diverse cardiac abnormalities. Despite its ubiquity, manual interpretation remains time-consuming, labor-intensive, and prone to significant inter-observer variability—even among experienced cardiologists [2]. This subjectivity, compounded by the exponential growth of ECG data, highlights the pressing need for automated systems that deliver accurate, consistent, and timely diagnoses.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have demonstrated remarkable performance in ECG classification, often rivaling expert-level accuracy [3]. However, as these models become increasingly complex, their decision-making processes grow opaque. This "black-box" nature undermines trust, accountability, and clinical adoption, since healthcare professionals are understandably reluctant to rely on predictions that lack transparent reasoning.

A growing body of work has attempted to bridge this gap through post-hoc explanation techniques such as Grad-CAM [4]. Yet these methods remain fundamentally limited: they infer rationales retrospectively, often producing coarse or unstable attributions that fail to capture the nuanced diagnostic patterns the model actually exploits. This suggests that genuine interpretability should be a fundamental component of the model architecture itself, rather than a post-hoc add-on. Building on this principle, our central hypothesis is that complex ECG diagnosis—whether involving co-occurring pathologies (multi-label) or mutually exclusive arrhythmias (multi-class)—can be more effectively addressed by decomposing the problem into a set of parallel, specialized tasks. By coupling each task with an intrinsically interpretable module, the model can simultaneously achieve high diagnostic accuracy and produce clinically meaningful explanations.

To this end, we propose **MeDeA (Multi-disease Decompositional Attention)**, a novel architecture for intrinsically interpretable ECG classification. Our contributions are as follows:

ORCID(s):

- **A decomposable attention framework** that reframes ECG diagnosis by assigning each disease category to a dedicated Multi-Query Attention Head. This design supports both multi-label and multi-class paradigms, enabling the learning of disentangled, class-specific diagnostic criteria.
- **The Multi-Query Attention Head (MQAH) design**, conceptualized as an "expert committee" that captures multiple discriminative waveform patterns for a single disease, effectively mitigating the "winner-takes-all" feature collapse observed in conventional attention mechanisms.
- **Extensive empirical validation** on two diverse benchmarks: the multi-label PTB-XL dataset and the single-label Chapman dataset. Results demonstrate that MeDeA achieves state-of-the-art performance while maintaining intrinsic interpretability aligned with clinical reasoning.
- **A systematic attribution framework** for visualizing class-specific attention maps, providing transparent insight into the model's internal reasoning and facilitating trustworthy human-AI collaboration.

The remainder of this paper is organized as follows. Section II reviews related work in ECG analysis and explainability. Section III details the MeDeA architecture and its task-adaptive loss strategy. Section IV reports our experimental setup and comprehensive results, and Section V concludes the paper.

2. Related Work

2.1. Deep Learning for ECG Classification

The evolution of deep learning in ECG analysis has largely paralleled advancements in sequence modeling. Early approaches predominantly utilized Convolutional Neural Networks (CNNs) as feature extractors, leveraging architectures such as ResNet to capture local morphological patterns like P-waves and QRS complexes [5]. While effective for local feature extraction, CNNs are intrinsically limited by their receptive fields, struggling to capture long-range temporal dependencies essential for detecting paroxysmal arrhythmias.

To address temporal modeling, Recurrent Neural Networks (RNNs), particularly LSTMs and GRUs, were introduced to process sequential dependencies [6]. Hybrid architectures combining CNN encoders with RNN decoders further improved performance by integrating morphological and temporal features [7]. However, recurrent models often suffer from training inefficiencies and vanishing gradients when processing long-duration recordings.

Most recently, Transformer-based architectures have emerged as the state-of-the-art, utilizing self-attention to model global dependencies without the constraints of recurrence [8]. While studies indicate that attention-based models can outperform traditional temporal networks [9], the quadratic computational complexity of standard self-attention remains a bottleneck for high-sampling-rate physiological signals [10]. Furthermore, as highlighted in a recent systematic review by Wu and Guo (2025) [11], the pursuit of accuracy often comes at the cost of interpretability and generalization. Current research increasingly emphasizes the need for architectures that balance diagnostic precision with trustworthiness and computational feasibility in clinical settings.

2.2. Explainability in Medical AI

The opacity of deep learning models poses a significant barrier to clinical deployment, motivating research into two primary interpretability paradigms: **post-hoc explanation** and **intrinsic interpretability** [12].

Post-hoc methods, such as Grad-CAM [4], attempt to approximate the decision boundaries of a trained "black-box" model. While widely used, these methods are fundamentally retrospective approximations. Recent evaluations in time-series domains reveal that post-hoc attributions are often unstable and can produce conflicting explanations for the same prediction under minor perturbations [13], limiting their reliability for high-stakes medical diagnoses.

In contrast, intrinsic interpretability aims to embed transparency directly into the model structure. Attention mechanisms are a prominent example, as attention weights effectively serve as a built-in saliency map [14]. However, in multi-label or multi-class settings, standard shared attention mechanisms often suffer from **explanation entanglement**, where a single attention map vaguely aggregates features from multiple distinct pathologies, obscuring the specific evidence for each decision [15]. MeDeA addresses this by enforcing architectural decomposition, ensuring that evidence for each disease is isolated and verifiable.

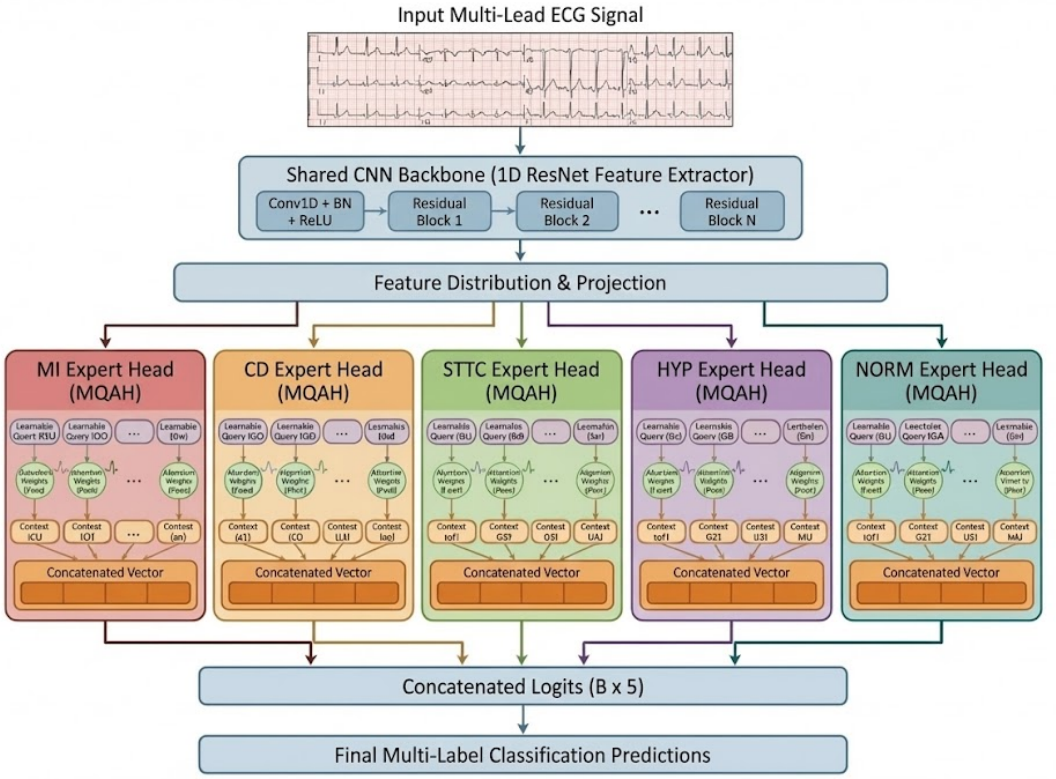

Figure 1: Schematic overview of the MeDeA architecture illustrating the top-down data flow. The pipeline begins with a raw multi-lead ECG signal processed by a **Shared 1D ResNet Backbone** to extract temporal features. These features are then decomposed and routed to a committee of five parallel, disease-specific **"Expert Heads"**. Each head operates independently to identify morphological evidence relevant to its specific pathology, outputting logits that are concatenated for the final multi-label diagnosis.

2.3. Attention Mechanisms and Feature Collapse

Standard Multi-Head Attention (MHA) extends self-attention by partitioning the representation subspace, allowing the model to jointly attend to information from different representation subspaces at different positions. While MHA improves expressivity, standard softmax-based attention often exhibits a "winner-takes-all" phenomenon [16]. This behavior causes probability mass to concentrate excessively on a single discriminative token, suppressing other relevant but less dominant features.

In the context of ECG diagnosis, this "feature collapse" is problematic because a robust diagnosis often relies on a constellation of morphological evidence (e.g., the co-occurrence of ST-segment elevation and pathological Q-waves) rather than a single point of interest. Unlike standard approaches that optimize for subspace diversity, our proposed method introduces a multi-query formulation within the same feature space. This design is explicitly motivated by the need to overcome the winner-takes-all limitation, acting as an ensemble of morphological probes to capture comprehensive clinical evidence.

3. Our Method

In this section, we introduce the architecture and design principles of our proposed model, **MeDeA (Multi-disease Decompositional Attention)**. The core philosophy relies on the premise that interpretability in complex diagnostic tasks is best achieved by structurally decomposing the problem into distinct, clinically meaningful sub-problems. We first outline the architectural overview, then detail its main components: the Shared CNN Backbone, the novel Multi-Query Attention Head, the task-adaptive loss strategy, and the mechanism for intrinsic explanation generation.

3.1. Architectural Overview

The MeDeA framework enforces a clear division between feature extraction and decision-making, embodying a "feature sharing, decision separation" paradigm [17]. As shown in Fig. 1, the pipeline begins with a raw ECG signal (single or multi-lead). This signal is first processed by a **Shared CNN Backbone**, which functions as a feature extractor, transforming the raw time-series into a sequence of high-level representations.

The extracted feature sequence is then fed in parallel to a set of independent **Multi-Query Attention Heads (MQAH)**, each dedicated to a specific disease category. Each head analyzes the full feature sequence to output both a diagnostic logit and an associated attention map, serving as an intrinsic explanation. The logits from all heads are concatenated to form the final prediction vector. Crucially, this decomposable design supports both multi-label (independent binary decisions) and multi-class (mutually exclusive) tasks, avoiding the feature entanglement often seen in monolithic classifiers [18].

3.2. Shared CNN Backbone

Raw physiological signals are high-dimensional and noisy, making direct application of attention mechanisms inefficient [19]. The Shared CNN Backbone acts as a feature engineering module, converting the long, noisy signal into a shorter, information-dense sequence.

We adopt a 1D ResNet-based architecture for this backbone [20]. It begins with an initial convolutional layer for downsampling, followed by a stack of residual blocks. This deep convolutional structure effectively learns hierarchical features, from local waveform slopes to complex morphological patterns of the P-QRS-T complex. The backbone outputs a feature sequence $X_{\text{feat}} \in \mathbb{R}^{B \times C \times L'}$, where B is the batch size, C the number of channels, and L' the reduced sequence length. This representation provides an abstract yet informative view of the ECG, optimized for subsequent diagnostic reasoning.

3.3. Multi-Query Attention Head

As shown in Fig. 2, the Multi-Query Attention Head (MQAH) is the core innovation of MeDeA, designed to yield clinically intuitive explanations.

Motivation A reliable clinical diagnosis often requires synthesizing multiple, spatially distant pieces of evidence (e.g., co-occurring ST-segment elevation and reciprocal depression). Standard softmax-based attention, however, exhibits a "winner-takes-all" behavior, concentrating weight on a single discriminative feature. This leads to the "single-point highlight" effect, limiting interpretive value.

Architectural Design To address this, we replace the single query vector with a committee of H learnable query vectors, $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_H\}$, each specializing in a different disease-specific feature pattern.Each query vector $\mathbf{q}_h$ is implicitly guided by the disease-specific supervision to focus on distinct morphological sub-patterns, thereby distributing the attention mass across multiple relevant evidence rather than collapsing to a single peak.

Given the backbone features, we project them into key ($\mathbf{K}$) and value ($\mathbf{V}$) spaces. Each query vector computes a separate attention distribution. To mitigate the concentration effect, we introduce a **temperature scaling** parameter $\tau > 1$ [21], yielding attention weights for the h-th query:

$$\boldsymbol{\alpha}_h = \text{softmax}\left(\frac{\mathbf{q}_h \mathbf{K}^T}{\sqrt{d_k} \cdot \tau}\right), \tag{1}$$

where d_k is the key dimension. The resulting context vector is $\mathbf{c}_h = \boldsymbol{\alpha}_h \mathbf{V}$. Finally, context vectors from all H queries are concatenated and passed through a feed-forward network (FFN) to produce the class logit:

$$\text{logit} = \text{FFN}(\text{Concat}(\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_H)). \tag{2}$$

This "expert committee" enforces multi-evidence reasoning, resulting in more robust predictions and richer explanations.

3.4. Loss Function

To ensure optimal performance across different diagnostic paradigms, we employ a **Task-Adaptive Loss Strategy**:

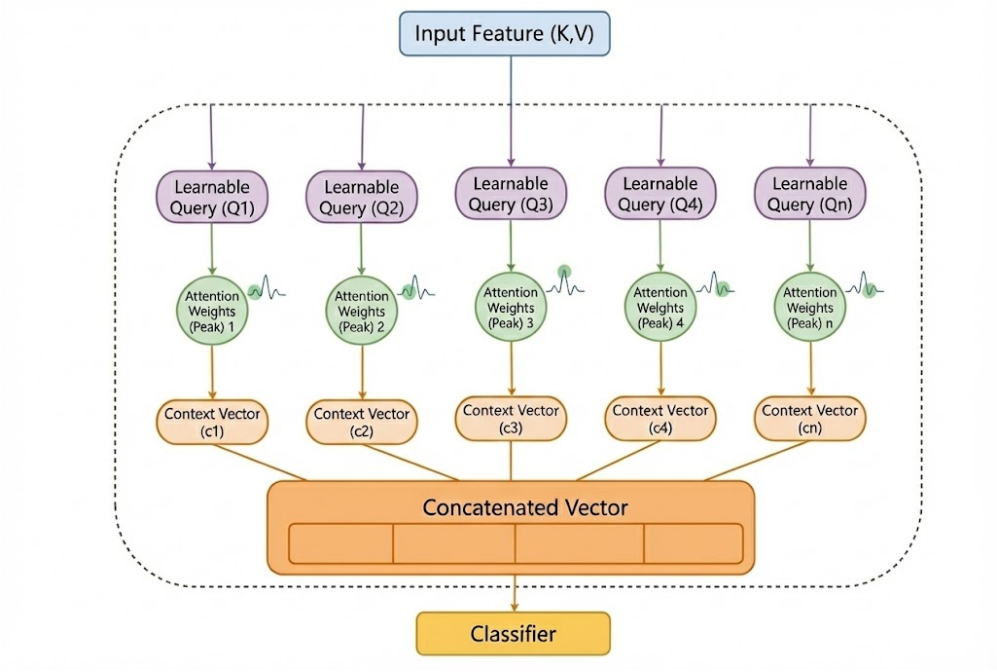

Figure 2: Micro-architecture of a single Disease-Specific Expert Head. This diagram details the internal mechanism of the Multi-Query Attention Head (MQAH). Within each expert, a set of **Learnable Queries** ($Q_1 \dots Q_n$) act as independent morphological probes. They interact with the input features (K, V) to generate distinct **Attention Weights** (visualized as localized signal peaks) and extract corresponding **Context Vectors** ($c_1 \dots c_n$). These vectors are aggregated to produce the final disease-specific logit.

3.4.1. Multi-Label Scenario

For datasets like PTB-XL where pathologies can co-occur, we utilize the **Binary Cross-Entropy with Logits Loss (BCE)**. This treats each disease class c as an independent Bernoulli distribution. Let $y_c \in \{0, 1\}$ be the ground truth and $\hat{y}_c$ be the predicted logit for class c. The loss is defined as:

$$\mathcal{L}_{\text{BCE}} = -\sum_{c=1}^{C} w_c \left[y_c \log(\sigma(\hat{y}_c)) + (1 - y_c) \log(1 - \sigma(\hat{y}_c)) \right] \tag{3}$$

where $\sigma(\cdot)$ is the sigmoid function and w_c is the positive class weight to handle imbalance.

3.4.2. Single-Label Scenario

For datasets like Chapman where classes are mutually exclusive, we employ the standard **Cross-Entropy Loss (CE)**:

$$\mathcal{L}_{\text{CE}} = -\sum_{c=1}^{C} y_c \log \left(\frac{\exp(\hat{y}_c)}{\sum_{j=1}^{C} \exp(\hat{y}_j)} \right) \tag{4}$$

This ensures the model selects a single dominant class via the Softmax function.

3.5. Intrinsic Explanation Generation

Each MQAH directly produces its own explanation as H attention weight vectors. To consolidate these into a human-readable form, we apply two steps:

1. **Averaging:** Compute the element-wise average of the maps:

$$\boldsymbol{\alpha}_{\text{avg}} = \frac{1}{H} \sum_{h=1}^{H} \boldsymbol{\alpha}_h, \tag{5}$$

representing the collective focus of all queries.

2. **Upsampling:** Since $\boldsymbol{\alpha}_{\text{avg}}$ corresponds to the downsampled feature length L', we upsample it to the original signal length L via linear interpolation.

The resulting heatmap is overlaid on the raw ECG, offering a faithful visualization of the model's diagnostic focus.

4. Experiments and Results

In this section, we present a comprehensive evaluation of MeDeA. We structure our analysis into three parts: first, we detail the experimental settings and implementation; second, we provide a rigorous quantitative evaluation against state-of-the-art benchmarks and analyze component contributions via ablation studies; finally, we demonstrate the model's explainability through qualitative visualization of clinical cases.

4.1. Experimental Settings

4.1.1. Dataset and Preprocessing

We evaluate our approach on two benchmarks representing distinct diagnostic paradigms. First, the **PTB-XL dataset** [22] is used as a **multi-label** benchmark. It contains 21,837 clinical 12-lead ECG recordings where multiple pathologies can co-occur. We adopt the standard 5-superclass split: Normal (NORM), Myocardial Infarction (MI), ST/T Change (STTC), Conduction Disturbance (CD), and Hypertension (HYP). Second, to assess performance on **single-label** classification, we use the **Chapman-Shaoxing (Chapman) dataset** [23], focusing on 5-class arrhythmia classification where classes are mutually exclusive. The class distributions are summarized in Table 1 and Table 2.

Table 1
PTB-XL Dataset Superclass Distribution.

Superclass	Number of Samples	Prevalence (%)
NORM	9528	43.6%
MI	5486	25.1%
STTC	5255	24.1%
CD	4907	22.5%
HYP	2658	12.2%

Table 2
Chapman Dataset Class Distribution

Class	Number of Samples	Prevalence (%)
SR	1362	26.8%
SB	2195	43.1%
ST	639	12.6%
AF+AFib	501	9.8%
AT+SVT	390	7.7%

4.1.2. Implementation Details

All models were implemented in PyTorch [24] and trained on a single NVIDIA RTX 3090 GPU using the AdamW optimizer [25] (LR=1×10^{-3}, weight decay=1×10^{-4}, batch size=32) for up to 100 epochs. We applied the task-adaptive loss strategy described in Section III. Specifically, we used **BCEWithLogitsLoss** for the multi-label PTB-XL task and **CrossEntropyLoss** for the single-label Chapman task. Model selection was strictly monitored based on the **validation Macro F1-score** to ensure balanced performance.

4.2. Quantitative Performance Evaluation

This section establishes the empirical effectiveness of MeDeA as a classifier through comparative and component-wise analysis.

4.2.1. Comparison with State-of-the-Art Methods

Our final model, **MeDeA (Multi-Query)**, achieves a Macro F1-score of **77.53%** on the PTB-XL test set. As shown in Table 3 (Part 1), MeDeA outperforms other interpretable approaches such as ECGNet [26] (75.30%) and IMLENet [27] (73.90%), while remaining highly competitive with black-box Transformers like MSW-TN [28]. Importantly, all models in Table 3 are compared under the rigorous **multi-label** setting. On the supplementary Chapman dataset (Table 4), MeDeA demonstrates excellent generalization, achieving a state-of-the-art F1-score of **0.988** and Accuracy of **0.988** on the single-label task.

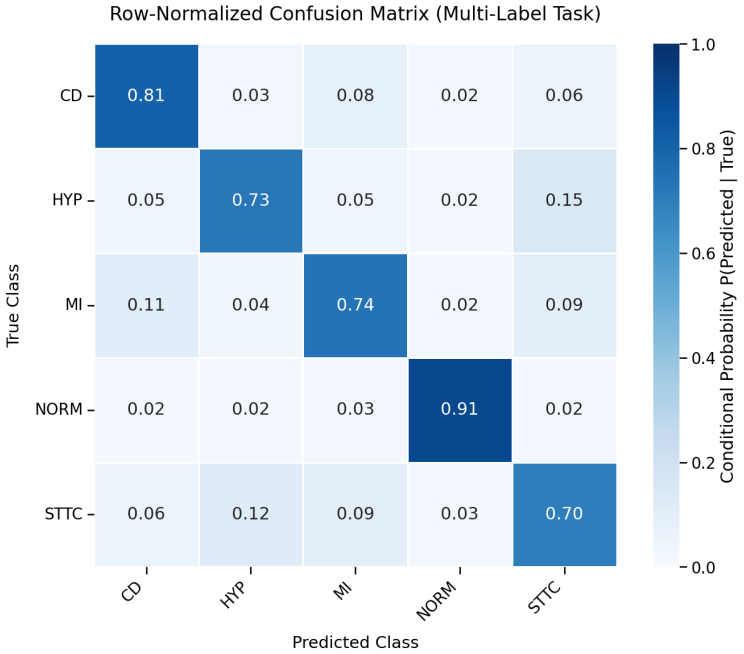

Figure 3: Row-normalized confusion matrix for the MeDeA model on the **PTB-XL (Multi-label)** task. While the task allows for co-occurring pathologies, this visualization highlights the model's effectiveness in identifying the dominant diagnostic category for each recording. The diagonal values represent the per-class recall, demonstrating consistent performance across all superclasses.

4.2.2. Per-Class Analysis and Confusion Patterns

Table 5 details the breakdown for each superclass in the PTB-XL dataset. The model maintains high AUC values (91.5%) across all conditions, including the minority HYP class. Additionally, to visualize the model's discriminative power, we present the confusion matrix for the **PTB-XL** dataset in Fig. 3. This matrix highlights the inter-class relationships and confirms that despite the multi-label nature, the model effectively distinguishes between primary pathologies.

4.2.3. Ablation Studies

We confirm the contribution of each architectural component in Table 6. Notably, removing the multi-query mechanism ('– w/ Single-Query') results in a clear performance drop, quantitatively validating our design choice before we explore its qualitative impact.

4.3. Qualitative Interpretability Analysis

In this section, we visually verify our core architectural claims by examining the model's reasoning process.

4.3.1. Clinical Consistency Verification

A trustworthy model must align with clinical knowledge. In cases of Myocardial Infarction (Fig. 4), MeDeA correctly focuses on pathological Q-waves and ST-segment abnormalities. Similarly, for Conduction Disturbance

Table 3
Comprehensive comparison on the PTB-XL dataset. Part 1 presents directly comparable results on the 5-superclass task (Macro F1). Part 2 summarizes recent representative works using alternative evaluation setups.

Model	Architecture / Key Idea	Setup	Score
Part 1: Direct Comparison on the 5-Superclass Task (Metric: Macro F1)			
MeDeA (Ours)	**CNN + Multi-Query Attn.**	**12L / 5 Cls.**	**0.775**
MSW-TN [28]	MS-Shifted Win. Transformer	12L / 5 Cls.	0.779
ResNet101 [22]	1D Residual Network	12L / 5 Cls.	0.762
HC-CNN-TN [29]	Hand-Crafted + CNN-Trans.	12L / 5 Cls.	0.760
ECGNet [26]	1D CNN + Bi-LSTM	12L / 5 Cls.	0.753
IMLENet [27]	Interpretable Multi-Level Net.	12L / 5 Cls.	0.739
Part 2: Broader Reference on Alternative Tasks (Metric: AUC)			
MeDeA (Ours)	**CNN + Multi-Query Attn.**	**12L / 5 Cls.**	**0.935**
xLSTM-ECG [30]	Feature Fusion with xLSTM	12L / 23 Cls.	0.911
Cardioformer [31]	Multi-Granularity Patch+ResNet	12L / 5 Cls.	0.900
KE-ECG [32]	Multimodal Knowl. Fusion	12L / 71 Cls.	0.890
Masked Train [33]	Robust Detection (Images)	12L / 23 Cls.	0.835

Table 4
Comprehensive comparison on Chapman arrhythmia classification tasks. We focus strictly on the 5-Class split to ensure a fair comparison with recent SOTA benchmarks.

Model	Architecture / Key Idea	Setup	Score
Part 1: Direct Comparison on the 5-Class Task (Metric: F1-score)			
MeDeA (Ours)	**CNN + Multi-Query Attn.**	**1L / 5 Cls.**	**0.988**
EXGnet-L [34]	XAI-Guided Multires. Net + Feats.	1L / 5 Cls.	0.981
EXGnet [34]	XAI-Guided Multires. Net	1L / 5 Cls.	0.979
xECGArch [35]	Trustworthy DL Arch.	1L / 5 Cls.	0.975
G2-ResNeXt [36]	ResNeXt w/ Group Conv.	1L / 5 Cls.	0.973
Part 2: Direct Comparison on the 5-Class Task (Metric: Accuracy)			
EXGnet-L [34]	XAI-Guided Multires. Net + Feats.	1L / 5 Cls.	0.989
MeDeA (Ours)	**CNN + Multi-Query Attn.**	**1L / 5 Cls.**	**0.988**
EXGnet [34]	XAI-Guided Multires. Net	1L / 5 Cls.	0.988
xECGArch [35]	Trustworthy DL Arch.	1L / 5 Cls.	0.986
G2-ResNeXt [36]	ResNeXt w/ Group Conv.	1L / 5 Cls.	0.984

(Fig. 5), the attribution map accurately highlights the widened QRS complex. These results demonstrate that MeDeA relies on clinically meaningful features rather than artifacts.

4.3.2. Impact of Multi-Query Mechanism

We compare MeDeA against a single-query baseline to validate our core innovation. As shown in Fig. 6, the single-query approach yields diffuse, unfocused attention. In contrast, the Multi-Query design (Fig. 7) enables the model to identify multiple, distinct morphological patterns, offering precise and actionable explanations.

Table 5
Per-Class Performance of MeDeA (Multi-Query).

Class	Precision	Recall	F1-Score	AUC
NORM	84.20%	90.10%	87.00%	95.70%
MI	82.60%	75.00%	78.60%	93.70%
STTC	77.00%	76.00%	76.50%	93.20%
CD	76.50%	84.00%	80.05%	93.30%
HYP	59.10%	73.50%	65.50%	91.50%

Table 6
Ablation study of the MeDeA model.

Model Configuration	Macro F1	Macro AUC	Experimental Purpose
Our Full Model (Baseline)	**77.53%**	**93.48%**	*(A) Complete model as baseline.*
– *w/ Single-Query*	74.82%	93.46%	(B) Test the necessity of multi-query.
– *Backbone Only (w/o Attention)*	73.89%	93.54%	(C) Assess the role of attention.
– *w/ Single-Head*	74.46%	93.39%	(D) Evaluate multi-head necessity.
– *w/ Fixed Query*	74.76%	93.67%	(E) Test the value of learned queries.
– *Attention Only (w/o CNN)*	61.50%	87.62%	(F) Test backbone's importance.
– *w/o Regularization*	73.24%	92.51%	(G) Assess effect of regularization.
– *w/ Simple CNN*	74.56%	93.27%	(H) Test backbone complexity.

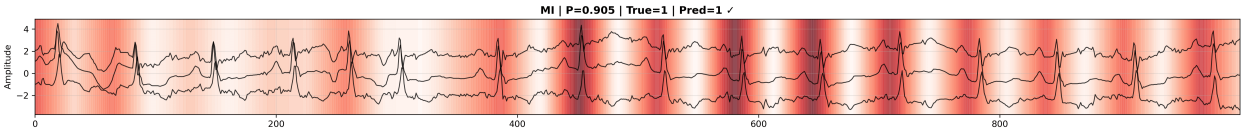

Figure 4: Attribution map for a patient with Myocardial Infarction (MI).

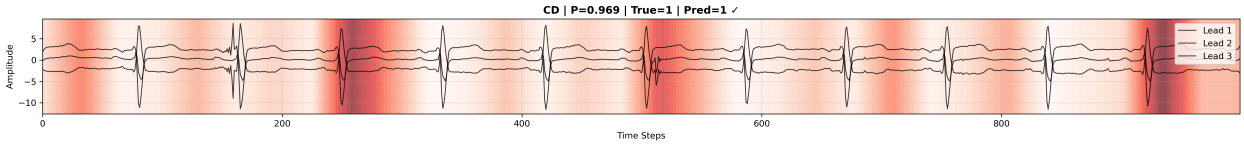

Figure 5: Attribution map for a patient with Conduction Disturbance (CD).

4.3.3. Decomposability in Multi-Label Cases

Fig. 8 demonstrates the model's capability in a complex case with three concurrent diagnoses (MI, STTC, CD). MeDeA generates distinct attribution maps for each condition, avoiding "explanation entanglement" and confirming the effectiveness of our decomposable architecture.

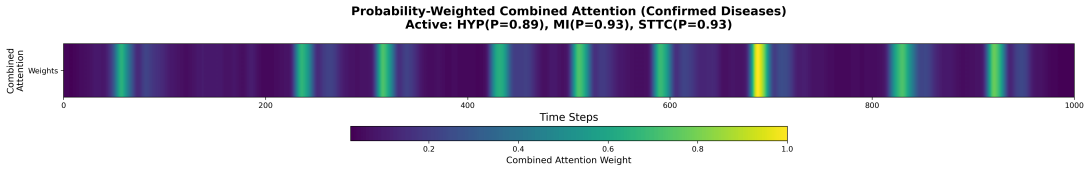

Figure 6: Explanation by baseline `MeDeA (Single-Query)`: diffuse and unfocused.

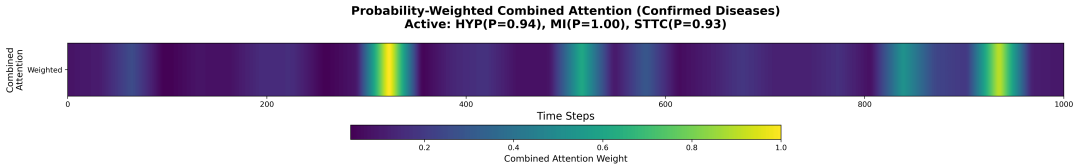

Figure 7: Explanation by `MeDeA (Multi-Query)`: precise and multi-focal.

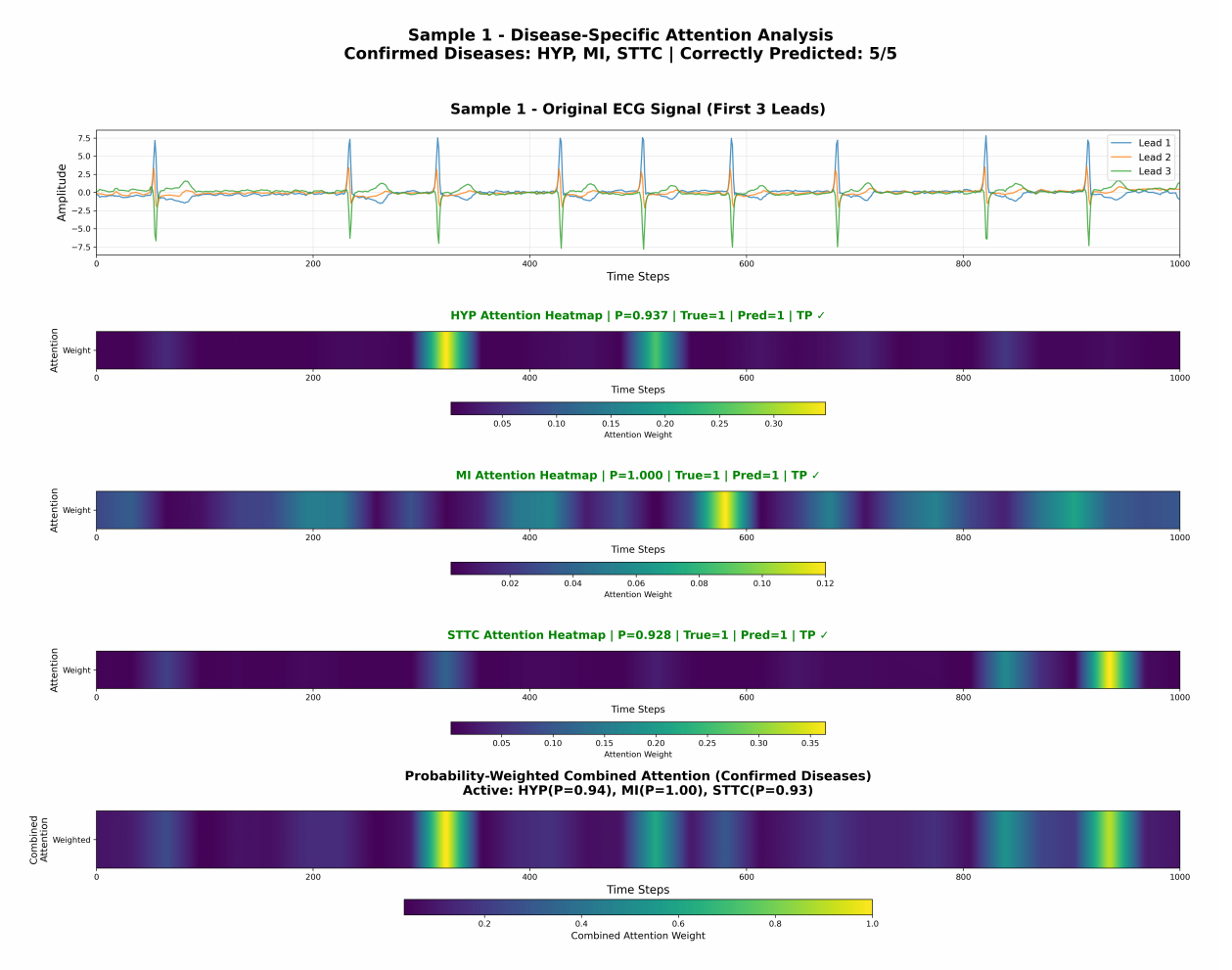

Figure 8: Explanation decomposability in a complex case (HYP, MI, STTC).

5. Conclusion

In this work, we addressed the critical "black-box" problem that hinders the clinical adoption of deep learning in cardiology. By adhering to the first principle that interpretability must be an intrinsic architectural property rather than a post-hoc add-on, we proposed **MeDeA**, a decompositional attention framework that reframes **complex** ECG diagnosis into a set of transparent, parallel sub-problems.

Our empirical evaluations rigorously demonstrate that MeDeA successfully resolves the tension between model performance and transparency. Quantitatively, the model achieves a Macro F1-score of **0.775** on the multi-label PTB-XL dataset. While this result is marginally lower than the state-of-the-art black-box transformer MSW-TN (0.779), it represents a highly favorable trade-off: MeDeA exchanges a negligible fraction of theoretical accuracy

for the indispensable clinical value of transparency—a capability entirely absent in black-box baselines. Furthermore, MeDeA outperforms existing interpretable methods (e.g., ECGNet at 0.753) and demonstrates robust generalization, establishing a new state-of-the-art F1-score of **0.988** on the single-label Chapman dataset. These results confirm that enforcing structural interpretability does not require a significant compromise in diagnostic accuracy.

Beyond numerical metrics, the core innovation of MeDeA lies in its **Multi-Query Attention Head (MQAH)** mechanism. Our qualitative analysis reveals that this design effectively overcomes the "winner-takes-all" feature collapse observed in standard attention mechanisms. By enabling the simultaneous localization of multiple, distinct pieces of morphological evidence (e.g., co-occurring ST-segment changes and Q-waves), MeDeA produces high-fidelity attribution maps that align with human clinical reasoning. This alignment supports a shift towards **collaborative diagnosis**, where clinicians can verify model predictions against highlighted morphological features, thereby filtering potential false positives and building trust in automated systems.

Despite these advancements, our study has limitations that outline directions for future research. First, our current evaluation is retrospective; prospective studies are required to quantify the actual impact of MeDeA's explanations on decision-making speed and accuracy in real-world clinical workflows. Second, the model relies on high-quality annotated data, and its robustness against significant domain shifts remains to be fully explored. Looking ahead, we aim to extend this decomposable framework to multi-modal settings or other physiological signals [37], integrating ECG with electronic health records or imaging data to further enhance the precision and holistic nature of automated cardiac diagnosis.

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